Week 4 Report

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Role: Frontend Developer & Scrum Master

Date: August 24, 2025

Project: OAS - Web-based Oral Assessments

Reporting Period: Monday, August 18, 2025 – Sunday, August 24, 2025

1. Tasks Completed This Week

Frontend Development & Backend Integration

Complete Password Reset System Integration (Evidence Link)

- **Spring Boot Backend Integration:**
  - Integrated with Spring Boot endpoints: POST /auth/password-reset/request and POST /auth/password-reset/confirm.
  - Implemented a two-step UI flow: forgot password → reset password with email verification.
  - Enhanced error handling for validation errors (code 4006, invalid codes, etc.).
  - Added React Error Boundary for graceful error handling.
- **Security & UX Enhancements:**
  - Implemented account enumeration prevention with generic success messages.
  - Added rate limiting and attempt tracking.
  - Included real-time password strength validation with visual feedback.
  - Ensured session management with automatic cleanup on successful reset.
  - Created user-friendly error messages without technical popups.

Onboarding System Backend Integration (Evidence Link)

- **Complete Backend Integration:**
  - Successfully integrated the onboarding system with the Spring Boot backend.
  - Updated the onboarding service to use real backend endpoints.
  - Enhanced the OnboardingWizard component for better backend compatibility.
  - Improved the password creation step with backend validation.
- **Service Layer Improvements:**
  - Refactored onboardingService.js with comprehensive backend integration.
  - Added proper error handling for backend responses.
  - Implemented token validation and user creation workflows.

Comprehensive User Management System (Evidence Link 1, Evidence Link 2)

- **Assignment-Centric User Management UI:**
  - Implemented the complete user management interface at /admin/users.
  - Created AddUserModal with role-based user creation.
  - Developed ScopeSelector for assignment and course context.
  - Built UsersTable with comprehensive user listing and actions.
  - Added ConfirmDialog for safe user deletion operations.
- **Service Integration:**
  - Created comprehensive userService.js with full CRUD operations.
  - Aligned API contracts with the backend implementation.
  - Added proper error handling and loading states.
  - Implemented search and filtering capabilities.

Examiner Self-Assignment System Design (Evidence Link)

- **User Flow & API Design:**
  - Designed a comprehensive examiner timeslot self-assignment flow.
  - Created detailed API contracts for timeslot management.
  - Documented a multi-step assignment process with availability checking.
  - Planned integration with calendar and notification systems.

Documentation & Architecture

Comprehensive API Documentation (Evidence Link)

- **User Management Documentation:**
  - Created detailed User-Management-Contracts.md with complete API specifications.
  - Developed User-Management-Flow.md with step-by-step user workflows.
  - Documented role-based access control and permission systems.
- **Project Structure Refinement:**
  - Restructured README.md with a clear architecture overview.
  - Updated documentation to reflect the separated frontend/backend architecture.
  - Added quick-start guides for the development setup.

Scrum Master Responsibilities

- **Sprint Coordination & Integration Management:** Coordinated the Week 4 sprint, facilitated integration testing sessions, and managed technical debt resolution.
- **Quality Assurance & Code Reviews:** Conducted comprehensive code reviews for all backend integration work and ensured architectural consistency.
- **Documentation Leadership:** Led documentation standardization efforts and created comprehensive API documentation for backend team collaboration.

2. Time Allocation (Approximation for Week 4)

- **Frontend Development:** 30 hours

    - Password Reset Integration: 8 hours
    - Onboarding System Integration: 6 hours
    - User Management System: 12 hours
    - Examiner Assignment Design: 4 hours
- **Scrum Master Activities:** 4 hours
    - Sprint coordination and integration planning: 2 hours
    - Code reviews and technical guidance: 1 hour
    - Team communication and standups: 1 hour
- **Documentation & Architecture:** 6 hours
    - API documentation creation: 3 hours
    - Design flow documentation: 2 hours
    - README and project structure updates: 1 hour

Total: 40 hours

3. Key Issues Encountered & Resolutions/Mitigation

Backend Integration Complexity

- **Issue:** The password reset and onboarding systems required significant refactoring to work with Spring Boot's specific response formats and error handling.
- **Resolution:** Redesigned the service layer architecture to properly handle backend responses and implemented React Error Boundaries for graceful error handling.
- **Mitigation:** Created comprehensive error mapping documentation and established testing protocols for all backend integrations.

User Management System Architecture

- **Issue:** The complex user management requirements needed a careful balance between an assignment-centric UI and role-based access control.
- **Resolution:** Implemented a modular component architecture with a ScopeSelector for context management and comprehensive permission handling.
- **Mitigation:** Created detailed API contracts and user flow documentation to ensure backend alignment and future maintainability.

API Contract Alignment

- **Issue:** Initial API designs required refinement to match Spring Boot backend capabilities and constraints.
- **Resolution:** Iteratively refined API contracts through collaboration with the backend team, ensuring realistic and implementable specifications.
- **Mitigation:** Established regular API review sessions and created comprehensive documentation for all contract changes.

4. Plan for Next Week (Week 5: August 25 - August 31,

2025)

Priority 1: User Management System Completion

- Complete backend integration testing for user management CRUD operations.
- Implement real-time user search and filtering with backend APIs.
- Add bulk user operations (import, export, bulk assignment).
- Validate role-based access control across all user management features.

Priority 2: Examiner Self-Assignment Implementation

- Implement the examiner timeslot self-assignment UI based on completed design flows.
- Create calendar integration for availability management.
- Develop a conflict resolution system for overlapping assignments.

Priority 3: Student Dashboard Enhancement

- Complete the student dashboard implementation with backend integration.
- Implement real-time assessment scheduling and notifications.
- Add assessment history and results viewing capabilities.

Scrum Master Focus Areas

- Coordinate final sprint planning for the user management system deployment.
- Facilitate comprehensive integration testing with the backend team.
- Manage preparation for user acceptance testing sessions.